From JIRAM images to winds

A pedagogical companion to Ingersoll et al. (2022)

Methods notes for the `jiram_catalog` project

9 September 2026

Abstract

Cloud tracking estimates the displacement of a brightness pattern. Turning that estimate into an atmospheric measurement requires a coordinate system, a time interval, a definition of the scale being measured, and an error model. This note develops those connections from elementary examples, motivated by the north-polar JIRAM analysis of Ingersoll and colleagues. It then identifies which building blocks our repository already has and which proposed features would make the resulting science easier to assess.

How to read this note. Section 2 is a compact record of the publication's reported recipe. The mathematical development that follows is an original teaching reconstruction, not the authors' software documentation. Worked examples are synthetic. Repository observations and future proposals are explicitly labelled. No new velocity retrieval has been run for this document.

Contents

1 The measurement we are trying to make **3**

2 The published recipe, in one place **3**

3 First establish what a pixel means **3**

3.1 Geometry precedes tracking 3

3.2 A mosaic is not necessarily an instant 4

4 How a correlation tracker measures a displacement **4**

4.1 Turn a patch into a search problem 4

4.2 Why this belongs to the optical-flow family 5

5 Texture determines where a vector is credible **5**

6 Four spatial scales that should never share one label **6**

7 From velocity to vorticity and divergence **7**

7.1 Physical definitions . . . 7

7.2 Measure a finite-area quantity using its boundary . . . 7

7.3 Three fields that expose sign and scale errors . . . 8

8 Why derivative uncertainty needs its own analysis **8**

9 Use repeated measurements to separate signal and noise **9**

9.1 Why four image sets are valuable . . . 9

9.2 Derive the covariance estimator . . . 9

9.3 What disjoint inputs do not guarantee . . . 10

10 What physical conclusions require beyond image tracking **10**

10.1 Shielding is a velocity-gradient statement . . . 10

10.2 Divergence is not a direct vertical-velocity measurement . . . 10

11 Where our repository already stands **11**

11.1 A usable baseline exists, with measurable limitations . . . 11

11.2 Three provenance details worth resolving before reproduction . . . 12

12 Features I would build from these lessons **12**

13 Sources, access limits, and how to reproduce this note **13**

Judgment calls and unresolved alternatives **14**

1 The measurement we are trying to make

Imagine two photographs of patterned fabric on a moving conveyor belt. Recognizing the same patch in both photographs gives a displacement. A ruler converts pixels into metres; a clock converts displacement into velocity. Cloud tracking has the same structure, with complications: the camera moves, the surface is curved, the texture changes, and the material can deform.

An image is a scalar field $I(\boldsymbol{x}, t)$. A horizontal velocity has two components, $\boldsymbol{v} = (u, v)$. The central inference is

$$I(\boldsymbol{x}, t) \;\longrightarrow\; \widehat{\boldsymbol{d}}(\boldsymbol{x}; t_0, t_1) \;\longrightarrow\; \widehat{\boldsymbol{v}}(\boldsymbol{x}) \;\longrightarrow\; \widehat{\zeta}, \widehat{D}, \tag{1}$$

where hats indicate estimates, ζ denotes relative vorticity, and D denotes horizontal divergence. Each arrow adds a requirement. In particular, the last arrow differentiates velocity and magnifies spatially varying errors.

Brightness patterns are tracers, not tagged air parcels. Even perfectly measured pattern motion can differ from material motion when clouds form, evaporate, change optical properties, or contain propagating waves. A useful workflow therefore keeps *image displacement*, *inferred wind*, and *dynamical interpretation* as distinct products.

2 The published recipe, in one place

The paper is *Vorticity and divergence at scales down to 200 km within and around the polar cyclones of Jupiter*, in *Nature Astronomy*, not *Nature Astrophysics* [1].

Element	Reported choice
Observations	JIRAM M-band, 2 February 2017; four sets of twelve images.
Mapping	SPICE/instrument geometry; polar orthographic plane; 15 km pixels.
Pairing	n01–n03 and n02–n04; approximately 16 minutes per pair, with actual image times; estimates approximately 8 minutes apart.
Tracker	JPL Tracker3; brightness correlation in 15×15 pixel patches (225 km sides).
Velocity sampling	45 km; native image sampling changes from 22 to 14 km.
Quality	Mask low-entropy patches; 256 brightness bins per patch.
Derivatives	Circulation and outward-flux integrals; square sides 90, 180, 270, 360 km.
Noise assessment	Compare disjoint image-pair estimates; fitted velocity-component uncertainty about $7.8\,\mathrm{m\,s^{-1}}$.

At the 180 km diagnostic scale, repeat-estimate correlations were about 0.729 for vorticity and 0.299 for divergence. The analysis supported anticyclonic shielding but did not detect the proposed divergence–anticyclonic vorticity correlation at its resolved scales. See Methods, Figs. 2–3, and Table 1 of the article [1].

3 First establish what a pixel means

3.1 Geometry precedes tracking

For a detector pixel, a geometric mapper uses the instrument model, spacecraft attitude and position, time, and a reference planetary surface to construct a viewing ray and its surface intersection.

A projection then places that intersection on a common map. This removes predicted camera motion before cloud motion is estimated. Reprojection must preserve missing coverage as a mask; filling a gap with black creates an artificial feature.

Let a map have coordinates (x, y) in metres. For a locally Cartesian grid, with pixel spacings s_x, s_y, a measured displacement (d_x, d_y) in pixels becomes

$$\widehat{u} = \frac{s_x d_x}{\Delta t}, \qquad \widehat{v} = \frac{s_y d_y}{\Delta t}, \qquad \Delta t = t_1 - t_0. \tag{2}$$

An array row is not automatically geographic north. Its direction can even oppose the map's positive vertical axis. A polar map also rotates the local east/north basis as longitude changes.

For an arbitrary projection, write the local relationship as

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathsf{J}(\lambda, \phi) \begin{pmatrix} u_E \\ v_N \end{pmatrix}, \tag{3}$$

where J is the projection Jacobian with respect to physical east and north distances. Invert this relationship to report east/north winds where the map is nonsingular. A simple map-plane derivative is a local approximation to a surface derivative. On a wide polar domain the metric must be included, or its neglected error quantified. An orthographic map preserves neither all distances nor all areas.

3.2 A mosaic is not necessarily an instant

A tiled image can contain observations acquired at different times. Preserve the source identity and acquisition time behind each tracked patch. When two patches use different source images, their Δt can differ even if both belong to the same nominal pair of mosaics. A cumulative display mosaic can also repeat old pixels; repeated display frames do not constitute independent exposures.

Worked example: units and navigation bias

Choose a synthetic grid with $s_x = s_y = 20{,}000$ m and $\Delta t = 900$ s. A displacement $(2.4, -1.2)$ pixels gives $(u, v) = (53.33, -26.67)\,\mathrm{m\,s^{-1}}$. A relative navigation offset of just 0.5 pixel adds $11.11\,\mathrm{m\,s^{-1}}$ along the affected axis. Correlation cannot distinguish that offset from a uniform translation of the clouds without independent geometric information.

4 How a correlation tracker measures a displacement

4.1 Turn a patch into a search problem

Choose a template patch A_i in the first image. For each trial displacement $\boldsymbol{d}$, compare it with a patch $B_i(\boldsymbol{d})$ in the second. A standard, useful realization is normalized cross-correlation:

$$C(\boldsymbol{d}) = \frac{\sum_{i \in M(\boldsymbol{d})} (A_i - \bar{A})(B_i(\boldsymbol{d}) - \bar{B})}{\left[\sum_{i \in M(\boldsymbol{d})} (A_i - \bar{A})^2 \sum_{i \in M(\boldsymbol{d})} (B_i(\boldsymbol{d}) - \bar{B})^2\right]^{1/2}}. \tag{4}$$

The means use the same jointly valid set $M(\boldsymbol{d})$ as the sums. The mask changes with displacement. Require adequate support and nonzero variance; otherwise the score can be undefined or based on too few samples.

The selected displacement maximizes C. Subtracting the mean makes the score insensitive to a constant brightness offset; dividing by the standard deviations removes a positive multiplicative gain. Neither operation corrects changing shadows, local radiometric distortion, or genuine cloud evolution.

An integer search can be refined near its maximum. For example, fit $q(\boldsymbol{\epsilon}) = a + \boldsymbol{b}^T\boldsymbol{\epsilon} + \frac{1}{2}\boldsymbol{\epsilon}^T \mathsf{H}\boldsymbol{\epsilon}$ to nearby correlation scores. If H is negative definite and the fitted maximum is nearby, then $\boldsymbol{\epsilon}_* = -\mathsf{H}^{-1}\boldsymbol{b}$ provides a subpixel correction. This is interpolation of a measured peak, not evidence for arbitrarily fine spatial resolution.

Equation (4) and this refinement describe the local reimplementation inspected for this note [3]. They are a concrete teaching model; the publication does not establish that every normalization, peak-fitting, or rejection choice is identical in its tracker.

4.2 Why this belongs to the optical-flow family

The simplest brightness-advection model is

$$I(\boldsymbol{x} + \boldsymbol{v}\Delta t, t + \Delta t) \simeq I(\boldsymbol{x}, t). \tag{5}$$

A first-order expansion gives

$$I_t + uI_x + vI_y \simeq 0. \tag{6}$$

There is only one scalar equation for two unknown components at a pixel. This is the aperture problem: a straight stripe moving along itself is locally indistinguishable from a stationary stripe. A textured patch provides gradients in several directions, making a common local displacement estimable.

Patch correlation searches finite displacements. A differential optical-flow method solves a version of Eq. (6), often with neighborhood or smoothness assumptions. A learned model obtains additional constraints from training data and its architecture. These are different ways to resolve ambiguity; a dense output array alone does not show which displacements were well constrained by the images.

More generally the image can obey

$$I_t + \boldsymbol{v} \cdot \nabla I = S_I, \tag{7}$$

with a source term S_I representing changing tracer brightness. Setting $S_I = 0$ is an assumption. A model may fit the images very well while assigning some of S_I to a spurious velocity.

5 Texture determines where a vector is credible

Shannon entropy of a patch histogram is

$$H = -\sum_{k:p_k>0} p_k \log_2 p_k. \tag{8}$$

If all samples occupy one bin, $H = 0$. If N samples occupy distinct bins, $H = \log_2 N$, provided at least N bins are available. This counts occupied intensity states, not spatial arrangements.

Consequently a monotonic ramp can have a broad histogram while constraining motion in only one direction. Random detector noise can have high entropy without persistent texture. Histogram entropy is useful as one diagnostic, but it is not a probability that a vector is correct. Its numerical threshold depends on binning, intensity scaling, and patch size.

A complementary mathematical diagnostic is the structure tensor

$$\mathsf{G} = \sum_i w_i \begin{pmatrix} I_{x,i}^2 & I_{x,i}I_{y,i} \\ I_{x,i}I_{y,i} & I_{y,i}^2 \end{pmatrix}. \tag{9}$$

Two appreciable eigenvalues imply sensitivity to both displacement components. One very small eigenvalue indicates an aperture ambiguity. This is a suggested additional check, not a reported ingredient of the publication's workflow.

Other useful proposed diagnostics are a distinct correlation peak, distance from the search boundary, common valid area, and forward/backward agreement. For the last check, let $\boldsymbol{p}$ and both displacements be in pixel units, and evaluate

$$\boldsymbol{r}_{\mathrm{FB}}(\boldsymbol{p}) = \boldsymbol{d}_{01}(\boldsymbol{p}) + \boldsymbol{d}_{10}(\boldsymbol{p} + \boldsymbol{d}_{01}(\boldsymbol{p})). \tag{10}$$

Consistent correspondence gives a small residual. It still cannot rule out a systematic error common to both directions.

6 Four spatial scales that should never share one label

1. **Native image sampling:** the physical distance between detector samples on the planet. It changes with viewing geometry.
2. **Map pixel spacing:** the interval of the resampled coordinate grid. Making this smaller creates additional interpolated samples.
3. **Template support and vector spacing:** a velocity uses texture throughout its patch; neighboring vector locations may use heavily overlapping patches. Support and spacing are separate quantities.
4. **Derivative scale:** the area or stencil over which velocity variations are converted into vorticity or divergence.

The values for the experiment are collected in Section 2. Its template width is five output-grid intervals. That ratio immediately warns against treating neighboring vectors as independent measurements. A half-width convention for wind localization should not be mistaken for a measured Fourier transfer function.

An oversampling thought experiment

Take a synthetic image whose useful texture exists only in 240 km patches. Evaluate a translation estimate every 40 km, then repeat every 10 km. The second display has sixteen times as many arrows per unit area. It has acquired no new photons and need not resolve any smaller eddies. It can still be useful for rendering and numerical integration, provided its dependence between neighboring estimates is retained.

One approximate mental model is

$$\widehat{\boldsymbol{v}} \simeq K * \boldsymbol{v} + \boldsymbol{e}, \tag{11}$$

where K represents spatial averaging and $\boldsymbol{e}$ error. A correlation estimator is nonlinear, so K is generally texture- and flow-dependent; it is not automatically a square top-hat the size of the template. Synthetic advection with known velocity is a way to measure the effective response.

7 From velocity to vorticity and divergence

7.1 Physical definitions

On a Cartesian tangent plane with x rightwards, y upwards, and positive normal out of the page,

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}, \qquad D = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}. \tag{12}$$

Positive ζ is counterclockwise rotation. Positive D means horizontal outflow from a small region. Both have units s^{-1}.

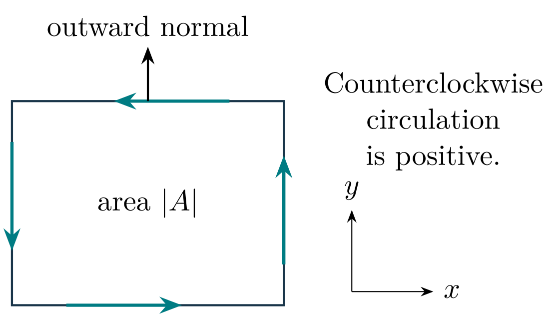
7.2 Measure a finite-area quantity using its boundary

For a region A with counterclockwise boundary ∂A, Stokes's and the planar divergence theorems give

$$\bar{\zeta}_A = \frac{1}{|A|} \oint_{\partial A} (u\,dx + v\,dy), \tag{13}$$

$$\bar{D}_A = \frac{1}{|A|} \oint_{\partial A} (u\,dy - v\,dx). \tag{14}$$

The first integral measures circulation along the boundary. The second measures flow across it, positive outwards. Division by area converts these integrated quantities into area-mean derivatives. These equations do not recover a point value at unlimited resolution.



For a square of side L, a transparent approximation using edge-mean velocities is

$$\bar{\zeta} \simeq \frac{\bar{v}_R - \bar{v}_L - \bar{u}_T + \bar{u}_B}{L}, \tag{15}$$

$$\bar{D} \simeq \frac{\bar{u}_R - \bar{u}_L + \bar{v}_T - \bar{v}_B}{L}. \tag{16}$$

Here R, L, T, B denote right, left, top, and bottom edges; the symbol L in the denominator denotes length. These are teaching approximations, not a claim about the authors' exact quadrature or interpolation rules.

An implementation should evaluate velocities along the complete supported contour. Replacing a missing edge by zero invents a circulation or flux. Interpolation across a small gap needs an explicit rule and uncertainty; unsupported contours should otherwise remain masked.

7.3 Three fields that expose sign and scale errors

Field	Velocity	Vorticity	Divergence
Translation	(U, V)	0	0
Rigid rotation	$(-\omega y, \omega x)$	2ω	0
Isotropic expansion	$(\alpha x, \alpha y)$	0	2α

These exact identities provide independent checks for either finite differences or contour integration. They also show why a spatially uniform navigation shift biases velocity but leaves ideal derivatives unchanged, whereas a rotation, scale change, or discontinuous tile offset can create false vorticity or divergence.

8 Why derivative uncertainty needs its own analysis

If each image's position estimate has independent component uncertainty σ_x in metres, then differencing two positions gives

$$\sigma_u = \frac{\sqrt{2}\,\sigma_x}{\Delta t}. \tag{17}$$

Longer baselines reduce this localization contribution while increasing the opportunity for deformation and tracer evolution. There is therefore an observational optimum, not a rule that the longest baseline is best.

If the four relevant edge-mean components in the square approximation have independent, equal uncertainty σ_e, then

$$\sigma_{\bar{D}} = \sigma_{\bar{\zeta}} = \frac{2\sigma_e}{L}. \tag{18}$$

This illustrative formula applies to *edge means*; substituting a single-vector uncertainty silently assumes those edge means have that same uncertainty. Overlapping templates and reused contour samples normally introduce covariance.

For any linear discretized diagnostic $q = \boldsymbol{w}^T \boldsymbol{z}$, the general propagation rule is

$$\mathrm{Var}(q) = \boldsymbol{w}^T \mathsf{C} \boldsymbol{w}, \tag{19}$$

where C is the covariance of the velocity samples. Oversampling does not justify setting the off-diagonal entries to zero.

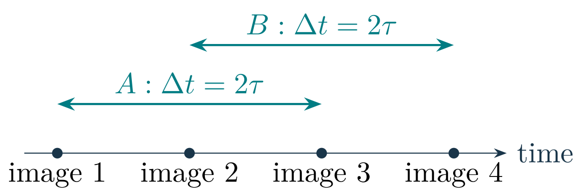
Worked example: a plausible wind error can dominate divergence

On the synthetic grid used earlier, 0.2-pixel position uncertainty in each image gives $\sigma_u = 6.29\,\mathrm{m\,s^{-1}}$. If, purely for illustration, each of four independent edge means has this uncertainty, a 200 km square gives $\sigma_{\bar{D}} = 6.29 \times 10^{-5}\,\mathrm{s^{-1}}$. A true divergence of $2 \times 10^{-5}\,\mathrm{s^{-1}}$ is then smaller than the noise in one estimate, even though a $60\,\mathrm{m\,s^{-1}}$ wind may look convincing. The edge-independence assumption is part of this example, not a calibration for Jupiter.

9 Use repeated measurements to separate signal and noise

9.1 Why four image sets are valuable

Consider equally spaced synthetic observations at $t_0, t_0 + \tau, t_0 + 2\tau, t_0 + 3\tau$. Form velocities from images $1 \to 3$ and $2 \to 4$:



The two estimates have equal baselines, midpoints separated by τ, and no input image in common. If structures change little over τ, the estimates can be treated approximately as repeat measurements of one field. This is a physical assumption to assess at the chosen diagnostic scale.

9.2 Derive the covariance estimator

Let $A = s + e_A$ and $B = s + e_B$ be two mean-subtracted estimates of a diagnostic. Assume errors are uncorrelated with s, uncorrelated with each other, and have equal variance σ_n^2. Then

$$\mathrm{Var}(A) = \mathrm{Var}(B) = \sigma_s^2 + \sigma_n^2, \tag{20}$$

$$\mathrm{Cov}(A, B) = \sigma_s^2, \tag{21}$$

$$\eta = \mathrm{corr}(A, B) = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_n^2}. \tag{22}$$

Thus η is the signal fraction of *variance* under this model, not a fraction of pixels that are correct and not a fraction of amplitude.

Another original way to see the same algebra is to form $S = (A+B)/2$ and $N = (A-B)/2$. Under those assumptions,

$$\mathrm{Var}(N) = \frac{\sigma_n^2}{2}, \qquad \mathrm{Var}(S) = \sigma_s^2 + \frac{\sigma_n^2}{2}, \qquad \sigma_s^2 = \mathrm{Var}(S) - \mathrm{Var}(N). \tag{23}$$

The difference estimates the noise of the *averaged* measurement. For unequal individual noise levels, $\mathrm{Var}(A) - \mathrm{Cov}(A, B)$ and $\mathrm{Var}(B) - \mathrm{Cov}(A, B)$ estimate their separate noise variances if the other assumptions still hold. Negative estimates in finite samples flag instability or model mismatch; they should not silently become evidence of no noise.

Worked example: interpret a correlation correctly

Suppose each measurement has standard deviation 10 in arbitrary diagnostic units and their correlation is 0.36. Their covariance is 36, so the common signal standard deviation is 6 and the individual noise standard deviation is 8. Averaging the pair reduces the independent noise standard deviation to $8/\sqrt{2} = 5.66$; the signal remains 6. These figures are synthetic.

9.3 What disjoint inputs do not guarantee

For evolving signals s_A, s_B, the covariance instead contains

$$\mathrm{Cov}(A, B) = \mathrm{Cov}(s_A, s_B) + \mathrm{Cov}(e_A, e_B) + \mathrm{Cov}(s_A, e_B) + \mathrm{Cov}(e_A, s_B). \tag{24}$$

Common navigation or calibration errors can survive in the covariance and masquerade as a repeatable signal. Real evolution can reduce signal covariance, making a discrepancy look like measurement noise. Correlation failures that depend systematically on the same texture can also recur.

Shared images cause an additional, explicit dependence. If image position errors p_1, p_2, p_3 are independent with variance σ_p^2, adjacent velocity errors are

$$e_{12} = \frac{p_2 - p_1}{\Delta t}, \qquad e_{23} = \frac{p_3 - p_2}{\Delta t}, \qquad \mathrm{Cov}(e_{12}, e_{23}) = -\frac{\sigma_p^2}{\Delta t^2}. \tag{25}$$

Pairs sharing an image are consequently not interchangeable with disjoint pairs in a noise calculation. Even the disjoint-pair calculation needs common spatial support, comparable temporal averaging, and uncertainty estimates that respect dependence among neighboring vectors.

10 What physical conclusions require beyond image tracking

10.1 Shielding is a velocity-gradient statement

For an ideal axisymmetric vortex with azimuthal wind $v_\theta(r)$,

$$\zeta(r) = \frac{1}{r}\frac{d(rv_\theta)}{dr}. \tag{26}$$

An annulus can therefore have negative vorticity even while $v_\theta > 0$: the velocity merely has to decrease faster than $1/r$. For the synthetic profile $v_\theta = Cr^{-p}$, $\zeta = (1-p)Cr^{-p-1}$, which is negative for $C > 0$ and $p > 1$. This explains how to interpret an anticyclonic shielding ring without requiring every wind arrow in that ring to reverse direction.

10.2 Divergence is not a direct vertical-velocity measurement

For a simple incompressible three-dimensional flow,

$$D = -\frac{\partial w}{\partial z}. \tag{27}$$

This constrains a vertical gradient, not w at a cloud level. A compressible atmosphere requires density-weighted continuity and information about vertical structure. Likewise, failure to detect a zero-lag correlation does not prove that a physical mechanism is absent: two causally related oscillatory signals can have zero correlation if one is a quarter cycle out of phase.

Siegelman et al. (2022) also analyzed these polar observations, combining tracked winds with infrared optical-depth anomalies and a surface quasigeostrophic interpretation to investigate smaller-scale convection and energy transfer [2]. Treat this as a distinct inference chain: the brightness field becomes a dynamical proxy under additional assumptions. Changing those assumptions can change the inferred divergence even when the input imagery is unchanged.

For this project, a direct velocity diagnostic and a model-assisted dynamical reconstruction should carry distinct names, assumptions, and uncertainty descriptions. Neither an attractive map nor a dense sampling grid establishes that a small-scale energy-transfer estimate is resolved.

11 Where our repository already stands

The following statements are observations from the local code, saved validation reports, and selected published-product headers; they are not additional results of the journal article.

11.1 A usable baseline exists, with measurable limitations

`src/jiram_catalog/tracking.py` implements masked normalized cross-correlation, subpixel peak refinement, and a reader for the published vector tables. Defaults include a 15-pixel square template, 35 trial-centre offsets per axis, at least 90% common valid support, and NCC at least 0.5. The 35 offsets span -17 to $+17$ pixels; this is our implementation's search convention. A peak at the search edge is retained without subpixel refinement. These choices do not establish identity with the original software [3].

The existing PJ4 report compares 24 corresponding same-letter tile pairs at subsampled published vector positions:

Input to our tracker	Median error	90th percentile	Within 1 pixel
Published maps	0.224 px	4.759 px	83.4%
Our reprojections	0.434 px	4.685 px	80.1%

The corresponding median vector-difference magnitudes are 3.45 and $6.69\,\mathrm{m\,s^{-1}}$. These are $\|\boldsymbol{v}_{\mathrm{ours}} - \boldsymbol{v}_{\mathrm{reference}}\|$, not differences between scalar speeds [4]. The medians provide encouraging agreement; the tails make clear that a single typical error cannot describe the whole field. The reference vectors are a published comparison standard, not an independent measurement of exact atmospheric truth. These saved results were inspected, not regenerated for this note.

The existing tracking gate checks saved summary metrics. It does not rerun all tracking or establish vorticity/divergence accuracy. The current GUI registration action estimates a bounded image displacement; its vector route loads an already associated vector product. Neither is a complete dense-wind retrieval and derivative-analysis workflow [5].

11.2 Three provenance details worth resolving before reproduction

1. **Tracker name.** The paper names Tracker3. The inspected `n0103aa.tp4` history contains `TASK='TRACKER4'`. Preserve both facts; source/version equivalence is unresolved.

2. **Vector sampling.** That raw table records `GRID=1`, `MPS=15`, and dense integer template positions. The published analysis spacing and the raw table's placement spacing are therefore different stages. The precise selection or aggregation producing the analysis grid needs verification. The folder name's "45 km" must not be interpreted as cloud altitude; an early repository inventory made that unsupported inference.

3. **Coordinates and clocks.** The example records `DT=974.002` s and sample/line displacement components. The repository reverses the line axis to match stored map rows. That reversal is empirically supported; the adopted one-based pixel origin is less strongly distinguished from a neighboring convention by tracking errors. Do not relabel these components as east/north without transformation. The repository's projection and region-array conventions also need a tested basis conversion: physical projection y points opposite to image line, whereas generic region y increases with row. A reflection can reverse the vorticity sign.

The published maps and vectors are under the read-only `JIRAM` reference directory configured by `JIRAM_PAPER_DATA`. They are separate from the working `jiram_mirror`. No files in either dataset were changed for this note.

12 Features I would build from these lessons

These are proposals for later specification and implementation. Their order reflects scientific dependencies.

1. **A four-set experiment builder.** Select disjoint pairs with comparable baselines and common coverage. Show every source timestamp, the effective temporal support, band, map basis, and instrument eligibility. Reject repeated cumulative pixels as independent exposures.

2. **A patch inspector.** Clicking an arrow should reveal the two patches, valid masks, correlation surface, subpixel fit, entropy, search boundary, and forward/backward residual. This makes a suspicious vector investigable instead of simply colouring it by a single score.

3. **An associated velocity product.** Save source IDs and versions, pair times, displacement, physical components, projection basis, masks, diagnostics, method parameters, and uncertainty provenance. Connect that product explicitly to the time-series viewer and existing vector overlay.

4. **Scale-aware derivative maps.** Offer circulation and flux diagnostics with the contour scale visible in kilometres. Display template support and vector spacing alongside it. Preserve unsupported contours and show noise estimates at each scale.

5. **A repeatability panel.** Compare disjoint-pair fields using scatterplots, covariance, average/difference maps, and sensitivity to scale and masking. Label the stationary-signal and error-independence assumptions. Use spatial blocks or independent realizations when estimating confidence intervals, rather than counting every overlapping vector as independent.

6. **A reproducible reference benchmark.** Begin with the published maps, then substitute our reprojections to isolate geometry contributions. Add synthetic translation, rotation, expansion, shear, missing coverage, brightness changes, and tile-offset cases. Compare the classical tracker with a proposed optical-flow model under matched sampling and filtering. Assess derivatives and effective resolution, not just displacement error.

JIRAM M-band thermal-emission/cloud-opacity texture and reflected-light JunoCam texture need different brightness-constancy checks. For JunoCam, keep the existing instrument-failure exclusions and native-band provenance. Timing, framelet assembly, limb/navigation residuals, and illumination changes need instrument-specific checks before transporting the JIRAM benchmark's confidence claims. A successfully exported tensor does not establish accurate winds. The most useful first milestone is a transparent PJ4 benchmark with associated vector products and scale-dependent uncertainty; generalizing to other passes then has an explicit standard.

13 Sources, access limits, and how to reproduce this note

The full main article, including Methods, was read from the coauthor-hosted PDF linked below. The publisher's supplementary-methods PDF could not be retrieved in this session. The numerical entropy cutoff, detailed contour quadrature, and any undocumented original Tracker settings are consequently not asserted here. Selected local supplementary-data headers and the repository's existing validation records provide the implementation evidence described in Section 11.

The document contains original schematics, derivations, and synthetic examples; it reproduces no article figures. With a standard TeX installation, compile from the repository root using:

```
pdflatex -output-directory=$TMPDIR
  docs/pedagogy/ingersoll_2022_tracking.tex
```

Run twice to resolve the table of contents and references. On Expanse, `TMPDIR` must point to the configured Lustre scratch runtime directory.

References

[1] A. P. Ingersoll et al. (2022). *Vorticity and divergence at scales down to 200 km within and around the polar cyclones of Jupiter*. Nature Astronomy **6**, 1280–1286. DOI: 10.1038/s41550-022-01774-0. Coauthor-hosted full text. Specific locators: Methods, pp. 1284–1285; Fig. 2 caption; Table 1 and Fig. 3; discussion of signal and noise, pp. 1281–1283.

[2] L. Siegelman et al. (2022). *Moist convection drives an upscale energy transfer at Jovian high latitudes*. Nature Physics **18**, 357–361. DOI: 10.1038/s41567-021-01458-y. See the discussion of optical-depth anomalies and the surface quasigeostrophic framework.

[3] Local repository: `src/jiram_catalog/tracking.py`. Inspected functions: `track_pair`, `parabolic_peak`, `read_tp4`, and `match_vectors`. Versioned source.

[4] Local repository: docs/reports/tracking_pj4.md; scripts/classical_tracking_pj4.py; tests/test_gate_tracking_pj4.py. Versioned validation report. The historical local_jiram_inventory.md contains superseded projection and altitude inferences; use the current geometry decisions and the source-qualified distinctions in this note.

[5] Local repository: src/jiram_catalog/api/science.py, docs/open_items.md, and docs/decisions.md. Versioned API source. Inspected 9 September 2026.

Judgment calls and unresolved alternatives

I chose a tutorial built around directly measured displacements and error propagation rather than a full reproduction of the paper's vortex-dynamics analysis. Normalized correlation and simple contour quadrature are explicit teaching realizations, not undocumented claims about Tracker3/4 internals. The numerical examples are illustrative rather than new Jupiter results.

I retained the distinctions between a map-plane vector and east/north wind, between raw table spacing and analysis spacing, and between published reference agreement and true atmospheric accuracy. The alternatives would be to infer those equivalences from filenames or good median errors; the available evidence does not support doing so. Exact software equivalence, raw-to-analysis vector aggregation, the entropy cutoff, and the original quadrature remain unresolved. The suggested application features require their own accepted implementation specification and scientific validation.